# Project Status Report 2

Project Name: Autonomous Reconnaissance and Evaluation System (ARES)

Team Name: JMAC Innovations

Project Manager: Pradhumna Shrestha

Team Members: Cynai Hayes, Marcus Lara Sanchez, Alexavier Mendez, Jackson Schroader

Report Date: 10/16/2026

Reporting Period: 9/28/2026 to 10/16/2026

## Management Summary

| | |
|---|---|
| Defined milestones completed: | 6 of 8 (75%) |
| Defined tasks completed: | 0 of 0 |
| Total estimated project hours used: | N/A |
| Ahead of (or Behind) schedule by: | On time |
| Known defects: | 0 open of 0 found |

## Schedule

| | |
|---|---|
| Initial estimated completion date: | *April 15, 2027* |
| Previous estimated completion date: | N/A |
| Current estimated completion date: | April 15, 2027 |

## Key Milestones Table

| ID | Title | Planned Completion Date | Previous Forecast Completion Date | Current Forecast Completion Date | Actual Completion Date |
|---|---|---|---|---|---|
| 1 | *Project Groups* | *8/20* | *N/A* | *N/A* | *8/20* |
| 2 | *Selected Projects* | *9/4* | *N/A* | *N/A* | 9/4 |
| 3 | *Wrote Requirements* | *9/24* | *N/A* | *N/A* | 9/24 |
| 4 | *Specifications Draft* | *9/28* | *N/A* | *N/A* | 9/28 |
| 5 | *Specifications Final* | *10/2* | *N/A* | *N/A* | 10/8 |
| 6 | *Preliminary Design* | *10/15* | *N/A* | *N/A* | 10/15 |
| 7 | *Project Plan/Detailed Design* | *10/22* | *N/A* | *10/22* | N/A |
| 8 | *Presentation Slides* | *11/28* | *N/A* | *11/28* | N/A |

## Product Size

*N/A*

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## Effort

| Life Cycle Activity | This Reporting Period (labor-hours) | | Project to Date (labor-hours) | |
|---|---|---|---|---|
| | Planned Effort | Actual Effort | Planned Effort | Actual Effort |
| GPS Testing for object detection to user | 8 | 9 | 9 | 10 |

## Cost

| Life Cycle Activity | This Reporting Period | | Project to Date | |
|---|---|---|---|---|
| | Planned Cost | Actual Cost | Planned Cost | Actual Cost |
| GPS Testing for object detection to user | 0 | 0 | 0 | 0 |

## Requirements Status

| ID | Functional Requirements | Team Member Responsible | Effort (%) | Completed (%) |
|---|---|---|---|---|
| FR 1 | The Robot Autonomous Operation in a field without user actively controlling it. | Marcus Lara Sanchez | 30 | 10 |
| FR2 | The system shall use the pre-designated field boundaries to determine a starting location and general scouting route on the field. | Alexavier Mnedez | 30 | 15 |
| FR3 | The system shall identify and follow crop rows while maintaining the intended scouting path during operation. | Jackson Schroader | 25 | 0 |
| FR4 | The system shall perform edge detection on captured image data to identify significant boundaries and features for further processing. | Jackson Schroader | 25 | 20 |
| FR5 | The system shall classify relevant field objects and hazards within its sensing area. | Cynai Hayes | 30 | 0 |
| FR6 | The system shall distinguish between desired crops and weeds for precise pesticide spraying | Jackson Schroader | 25 | 0 |
| FR7 | Object Localization to track robots location and location of detected objects | Marcus Lara Sanchez | 20 | 60 |
| FR8 | The system shall support a pre-scan mode for surveying a designated field area and recording detected objects | Alexavier Mendez | 30 | 0 |

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| | | | | |
|---|---|---|---|---|
| FR9 | Real Time Scout Mode where it scans for objects and pings the location of the objects/hazards. | Marcus Lara Sanchez | 15 | 60 |
| FR10 | The system shall record in real time the location and relevant information of detected objects to support the generation of a map for post-operation viewing and analysis. | Alexavier Mendez | 30 | 5 |
| FR11 | The system shall notify an operator of relevant hazards detected during real-time scouting | Cynai Hayes | 30 | 5 |
| FR12 | The system shall provide an interface through which an operator can configure scouting operations and review scouting results. | Cynai Hayes | 30 | 15 |
| FR13 | The system shall make stored results and identified objects available for later review, locally | Jackson Schroader | 10 | 0 |
| FR14 | System shall maintain lead distance ahead of agricultural equipment by regulating its speed along active path. | Marcus Lara Sanchez | 15 | 0 |

| ID | Non-Functional Requirements | Team Member Responsible | Effort (%) | Completed (%) |
|---|---|---|---|---|
| NFR1 | The system shall operate for a minimum of on a single battery charge under nominal scouting conditions. | Cynai Hayes | 5 | 0 |
| NFR2 | The system shall support an effective scouting width during normal field operation | Jackson Schroader | 5 | |
| NFR3 | Real Time Response where the system should detect a hazard and transmit an alert with an end to end latency of under 300ms. | Marcus Lara Sanchez | 20 | 10 |

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| | | | | |
|---|---|---|---|---|
| NFR4 | The FPGA-based classifier shall demonstrate improved computational performance relative to a CPU-based reference implementation, evaluated using metrics such as inference latency, throughput, power consumption, and energy per inference. | Jackson Schroader | 10 | 0 |
| NFR5 | The system shall support a repeatable live demonstration in a controlled environment that represents typical field conditions and allows the major scouting functions to be observed and evaluated. | Alexavier Mendez | 5 | 0 |
| NFR6 | The system shall maintain reliable operation and communication throughout a normal scouting mission without requiring manual restart or intervention. | Alexavier Mendez | 5 | 0 |
| NFR7 | The system shall have a robot to operate with | Cynai Hayes | 5 | 0 |

## Top Five Risks

1. *Cost – FPGAs are expensive so to tackle this finding a decently priced FPGA and researching how much resources we will utilize on it for our CNN model should help us with our budget.*
2. *Dust, mud, or sun blocking our sensor data. All of these factors can impact our sensors and even our camera. To fix this we can attach custom hoods to block from glare and debris that could interfere.*
3. *High latency could lead to hazards not being spotted in time. To mitigate the latency we will have to optimize our code as much as possible and experiment in different ways constantly to ensure we have a real time end to end communication.*
4. *The robot could stop in front of the machine. To prevent this and the risk of damage being done to both machines we will have to implement watchdog timers within our design so that our robot does not get stuck.*
5. *Communication between the user and robot. Once an object is detected the user is notified of where it is immediately. However, in most agricultural environments there is no cellular infrastructure to rely on so we will need to use our own wireless modules to send the hazard data.*

## Open Issues

1. *N/A*

## Action Items

*Pending Action Items:*

1. *N/A*

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*New Action Items:*

1. *Research parts and making parts list.*
2. *Collect data for rock edge detection*
    a. *Need a camera mount w/ similar angle to system for better training*
    b. *Will use video to capture many slices and angles of different rocks*
3. *Collect/Trim/Synthesize images for rock classification*
4. *Collect/Trim data for plant classification*

## Defects

1. *N/A*

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